Technical Report

1. Introduction

Business Analytics is the process of collecting, analyzing, and interpreting data to support better business decisions. Organizations generate large amounts of customer data every day, and analyzing this information helps identify trends, customer behavior, and business opportunities.

This project analyzes customer churn data to identify factors influencing customer retention. The study applies data cleaning, exploratory data analysis, descriptive statistics, correlation analysis, and hypothesis testing to generate actionable business recommendations.

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2. Business Problem

Customer churn has become a significant challenge for many organizations. Losing customers reduces revenue and increases the cost of acquiring new customers.

The company wants to understand:

- Why customers leave the service.
- Which customers are more likely to churn.
- How contract type influences customer retention.
- Whether monthly charges affect churn.
- Which payment methods are preferred.

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3. Project Objectives

The objectives of this project are:

- Clean and prepare customer data.
- Perform Exploratory Data Analysis (EDA).
- Identify patterns affecting customer churn.
- Apply statistical analysis.
- Perform hypothesis testing.
- Generate business recommendations.
- Develop an implementation plan.

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4. Dataset Description

The dataset contains more than 500 customer records.

Dataset Features

Feature	Description
CustomerID	Unique Customer ID
Tenure	Number of months customer stayed
MonthlyCharges	Monthly service charges
TotalCharges	Total amount paid
Contract	Contract type
PaymentMethod	Customer payment method
PaperlessBilling	Paperless billing status
SeniorCitizen	Senior citizen indicator

Feature	Description
Churn	Customer churn status

Dataset Type:

- Structured CSV File

Number of Records:

- 500+

5. Tools and Technologies

Programming Language:

- Python

Libraries Used:

- Pandas
- NumPy
- Matplotlib
- SciPy

Development Environment:

- Jupyter Notebook
- Visual Studio Code

Version Control:

- GitHub

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6. Data Cleaning and Preparation

Data preprocessing is one of the most important stages of analytics.

The following cleaning techniques were performed:

- Imported CSV dataset
- Checked dataset dimensions
- Verified data types
- Identified missing values
- Removed duplicate records
- Filled missing values
- Converted data types
- Saved cleaned dataset

Output:

- raw_data.csv
- cleaned_data.csv

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7. Exploratory Data Analysis (EDA)

EDA was performed to understand customer behavior and identify important trends.

The following visualizations were created:

- Customer Churn Bar Chart
- Contract Distribution Pie Chart
- Payment Method Bar Chart
- Monthly Charges Histogram
- Total Charges Histogram

- Customer Tenure Histogram
- Paperless Billing Pie Chart
- Senior Citizen Pie Chart
- Average Monthly Charges by Contract
- Correlation Matrix

Key Observations

- Month-to-Month contracts had higher churn.
- Long-term contracts retained more customers.
- Customers with higher monthly charges showed higher churn.
- Paperless billing adoption was high.
- Digital payment methods were widely used.

(Insert screenshots of your graphs here.)

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8. Statistical Analysis

The following descriptive statistics were calculated:

- Mean
- Median
- Standard Deviation
- Minimum
- Maximum

These measures helped summarize customer behavior and understand the distribution of numerical variables.

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9. Correlation Analysis

Correlation analysis was performed to examine relationships among numerical variables.

Variables analyzed:

- Tenure
- MonthlyCharges
- TotalCharges
- SeniorCitizen
- Churn

A correlation matrix was generated to visualize these relationships.

Findings

- Higher tenure was associated with higher total charges.
- Monthly charges showed a relationship with churn.
- Total charges increased as customer tenure increased.

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10. Hypothesis Testing

Objective

Determine whether Monthly Charges differ significantly between churned and non-churned customers.

Null Hypothesis (H_0)

There is no significant difference in Monthly Charges between churned and non-churned customers.

Alternative Hypothesis (H_1)

There is a significant difference in Monthly Charges between churned and non-churned customers.

Statistical Test

Independent Samples T-Test

Decision Rule

- If p-value < 0.05 → Reject H_0.
- If p-value ≥ 0.05 → Fail to Reject H_0.

Result

The statistical analysis indicated that Monthly Charges have a significant impact on customer churn.

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11. Business Insights

The analysis generated the following insights:

- Customers with Month-to-Month contracts are more likely to churn.
- Customers with long-term contracts are more loyal.
- Monthly charges influence customer retention.
- Total charges increase with longer customer relationships.
- Digital payment methods are popular among customers.
- Paperless billing is widely adopted.

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12. Business Recommendations

Based on the findings, the following recommendations are proposed:

1. Encourage customers to switch to long-term contracts.
2. Introduce customer loyalty programs.
3. Provide personalized retention offers.
4. Offer discounts for long-term subscriptions.
5. Improve customer support for high-risk customers.
6. Continue promoting digital payment methods.
7. Expand paperless billing adoption.
8. Regularly monitor customer churn trends.

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13. Business Implementation Plan

Phase	Duration	Activities
Phase 1	Month 1	Identify high-risk customers
Phase 2	Month 2	Launch retention campaigns
Phase 3	Month 3	Improve customer support
Phase 4	Month 4	Evaluate campaign effectiveness

Key Performance Indicators (KPIs)

- Customer Retention Rate
- Customer Churn Rate
- Revenue Growth
- Customer Satisfaction
- Customer Lifetime Value
- Average Monthly Revenue